Review retrospective: *the standard Gaussian moments* $(2k-1)!!$*, and an* $\mathbb{N}$*-subtraction at* $k=0$

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Abstract

A soundness review of `Laplace/OneD/GaussianMoments.lean`: the Stage-1 closed forms for the standard 1D Gaussian moments — $\int_{\mathbb{R}} x^{2k} e^{-x^2/2}\, dx = (2k-1)!!\sqrt{2\pi}$, its normalised and half-line versions, the odd-moment vanishing, and the rescaled $t^{-(k+1/2)}$ form. Result: **a clean fidelity pass with one landed edit** — a redundant `Gamma.Basic` import. The most interesting check was an $\mathbb{N}$-subtraction edge case at $k=0$, which turns out to be exactly right.

1 Setting

This is the primer's Stage 1 warm-up: the even moments of the standard Gaussian via the substitution $x = \sqrt{2u}$ to a Gamma integral, doubled by symmetry, with $\Gamma(k+1/2) = (2k-1)!!\sqrt{\pi}/2^k$ collapsing the prefactor to $\sqrt{2\pi}$. It is a foundational file, and the first review into the laplace 1D non-cumulant layer after the rescaling review.

2 What was reviewed

Five theorems: the half-line moment ($2^{k-1/2}\Gamma(k+1/2)$), the full-line moment ($(2k-1)!!\sqrt{2\pi}$), the normalised moment ($(2k-1)!!$), the odd-moment vanishing, and the rescaled moment ($(2k-1)!!\sqrt{2\pi}\, t^{-(k+1/2)}$ for $t>0$). The proofs lean on Mathlib's `integral_rpow_mul_exp_neg_mul_rpow`, `Real.Gamma_nat_add_half`, `integral_comp_abs`, and `integral_comp_mul_right`.

3 Fidelity / soundness findings

Sound; all five constants check out. The half-line constant $2^{k-1/2}\Gamma(k+1/2)$ matches the Mathlib master lemma at $p=2, q=2k, b=1/2$; the full-line collapse to $(2k-1)!!\sqrt{2\pi}$ is correct (the scalar $2\cdot 2^{k-1/2}/2^k = \sqrt{2}$ does the work); the rescaled exponent is exactly $t^{-(k+1/2)}$; and the odd moments vanish by the $\int f = \int f(-\cdot) = -\int f$ argument.

The check worth recording: the closed form is written as the double factorial of `(2 * k - 1)` with the subtraction in $\mathbb{N}$ (truncated). At $k=0$ this is `(2*0 - 1) = 0` in $\mathbb{N}$, so the right-hand side is $0!!=1$, and indeed $\int_{\mathbb{R}} x^0 e^{-x^2/2}\, dx = \sqrt{2\pi} = 1\cdot\sqrt{2\pi}$. So the $\mathbb{N}$-truncation accidentally lands on the correct value — it agrees with the mathematician's convention $(-1)!!=1$ — and for $k\ge 1$, $2k-1\ge 1$ so truncation never bites. GPT confirmed the edge case independently. This is the kind of thing that is easy to get subtly wrong in a Lean statement (a real $(2k-1)$ would go negative; the $\mathbb{N}$ version is fine precisely because $0!!=1$), so it was worth pinning down.

4 Simplifications applied

One landed: a redundant import. `Mathlib.Analysis.SpecialFunctions.Gamma.Basic` was imported explicitly, but `Real.Gamma` and `Real.Gamma_nat_add_half` arrive through `GaussianIntegral` (which the seabed's own API notes record as the home of `Gamma_nat_add_half`) and `Integral.Gamma`. Removed under rebuild-arbitration. Here the job count was telling in the *opposite* direction from the GaussianDomination case: it stayed at 2693 (unchanged), which means `Gamma.Basic` is still pulled in transitively through the other imports — so the explicit import was pure redundancy and, because the closure is unchanged, no downstream file can be affected. All five signatures byte-identical; GPT-affirmed; fast-forward-merged.

GPT's other proposals were declined: a genuinely-dead `rw [mem_Ioi] at hx` line (removing it would leave `hx` unused and trip the linter, requiring a coordinated binder rename — a proof-body edit for negligible benefit), and a long list of tactic-golf refactors (`simpa` collapses of the `rpow_natCast` rewrites, the scalar algebra, the even-symmetry step, the one-use `ring` helper, and the rescaling boilerplate). None removes dead code or changes a statement.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **ℕ-subtraction in a closed form is a fidelity trap worth a deliberate check — and here it passes.** `(2*k - 1)` in ℕ truncates to 0 at $k = 0$, and the proof is correct only because $0!! = 1$ coincides with the $(-1)!! = 1$ convention. When a paper formula has a subtraction that can underflow at the boundary, verify the boundary case explicitly rather than trusting the general form.
- **The job-count signal cuts both ways.** A redundant import can leave the count unchanged (still pulled in transitively, as here) or drop it (the import was the sole route, as in GaussianDomination). Either way the file's own green rebuild is the safety proof; the count just explains *why* it is safe — and an unchanged count is itself reassurance that no consumer's closure shrank.

7 Follow-ups

- The 1D non-cumulant layer continues: `Localisation.lean`, `Harmonic.lean`, the J-function and `Quartic/Sextic/Monomial` families are all tractable never-reviewed neighbours; the `Multi/Covariance*` trio still awaits a sectioned approach.